

AWS EC2 & Terraform – Step-by-Step Documentation

1. Introduction

This document explains how to:

- Launch an AWS EC2 instance **manually**
- Connect to the EC2 instance using **MobaXterm / PuTTY**
- Install and use **Terraform**
- Launch and delete an EC2 instance using **Terraform (Infrastructure as Code)**

This guide is written in **simple language** so that beginners can easily understand and follow each step.

2. Basic Concepts (Simple Explanation)

What is AWS?

AWS (Amazon Web Services) is a cloud platform that allows us to create servers, storage, and other services on the internet.

What is EC2?

EC2 (Elastic Compute Cloud) is a virtual server provided by AWS. We use EC2 to run applications or practice DevOps tasks.

3. Launching an EC2 Instance Manually (AWS Console)

Step 1: Login to AWS Console

- Open browser
- Go to <https://aws.amazon.com>
- Login to AWS Management Console

Step 2: Open EC2 Service

- Search for **EC2**
- Click on EC2 → EC2 Dashboard

Step 3: Launch Instance

- Click on **Launch Instance**
- Give an instance name (example: Manual-EC2)

Step 4: Choose AMI

- Select **Amazon Linux 2 / Amazon Linux 2023**
- This is free tier eligible and recommended

Step 5: Choose Instance Type

- Select **t2.micro** or **t3.micro**
- Free Tier eligible

Step 6: Create or Select Key Pair

- Choose **Create new key pair**
- Name it (example: ec2-key)
- Download the .pem file
- Save it safely (do not share)

Step 7: Configure Security Group

- Allow **SSH (Port 22)**
- Source: Anywhere or My IP

Step 8: Launch Instance

- Click **Launch Instance**
- Wait until instance state becomes **Running**

4. Connecting to EC2 Instance (MobaXterm / PuTTY)

4.1 Connecting Using MobaXterm (Recommended)

1. Open **MobaXterm**
2. Click **Session → SSH**
3. Enter:
 - Remote Host: **Public IP of EC2**
 - Username: **ec2-user**
4. Select **Use private key**
5. Choose your .pem key file

6. Click **OK**

You are now connected to the EC2 instance.

4.2 Connecting Using PuTTY (Windows)

1. Open **PuTTYgen**
2. Convert .pem file to .ppk
3. Open **PuTTY**
4. Enter Public IP in Host Name
5. Go to SSH → Auth → Browse and select .ppk file
6. Login user: ec2-user

What is Terraform?

Terraform is a tool that helps us create and manage cloud resources using code instead of doing everything manually.

5. Installing Terraform (Windows)

Step 1: Download Terraform

- Visit: <https://developer.hashicorp.com/terraform/downloads>
- Download Windows 64-bit version

Step 2: Install Terraform

- Extract zip file
- Copy terraform.exe
- Paste it in C:\Windows\System32 or add to PATH

Step 3: Verify Installation

Open Command Prompt:

```
terraform --version
```

If version appears, Terraform is installed successfully.

6. Launching EC2 Instance Using Terraform

Step 1: Create Project Folder

Create a folder:

```
terraform_ec2
```

Step 2: Create Terraform Files

provider.tf

```
provider "aws" {  
    region = "ap-south-1"  
}
```

variables.tf

```
variable "ami_id" {  
    description = "AMI ID"  
}
```

```
variable "instance_type" {  
    default = "t2.micro"  
}
```

```
variable "key_name" {  
    description = "EC2 Key Pair Name"  
}
```

main.tf

```
resource "aws_instance" "ec2_instance" {  
    ami      = var.ami_id  
    instance_type = var.instance_type  
    key_name  = var.key_name  
  
    tags = {
```

```
Name = "Terraform-EC2"

}

}
```

outputs.tf

```
output "public_ip" {

  value = aws_instance.ec2_instance.public_ip

}
```

Step 3: Run Terraform Commands

Initialize Terraform

terraform init

Create EC2 Instance

terraform apply

Type **yes** when asked.

Step 4: Verify EC2

- Go to AWS Console
- EC2 instance should be running
- Public IP displayed in Terraform output

7. Connecting to Terraform EC2 Instance

Use the same **MobaXterm** or **PuTTY** steps with:

- Public IP from Terraform output
- Same key pair

8. Deleting EC2 Instance (Cost Control)

Manual EC2 Deletion

- Go to EC2 Dashboard
- Select instance
- Click **Terminate**

Terraform EC2 Deletion

terraform destroy

Type **yes** to confirm.

9. Important Learnings

- Manual EC2 launch helps understand AWS basics
- Terraform automates infrastructure
- Infrastructure as Code is important for DevOps
- Always delete resources to avoid AWS charges
- SSH keys must be kept secure